On the Non-Invariance of Robustness Metrics: Baseline-Floor and Imputation Effects in Perturbed EEG Representations

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***Abstract*—Affective electroencephalographic (EEG) event-related potentials (ERPs) are noisy, subject-dependent projections of neural dynamics, and reported robustness rankings of EEG representations may depend more on *how* robustness is measured than on the representations themselves. We audit four affective-ERP representations spanning a fixed-to-trained spectrum—spectral band-power, ERP-window amplitudes, a fixed (untrained) leaky integrate-and-fire reservoir with a six-bin spike-count (BSC_6) embedding, and a trained EEGNet—under subject-grouped validation with amplitude-noise and channel-dropout perturbations, and identify two confounds that distort channel-dropout comparisons. First, an *accuracy-floor* confound: raw retention rewards near-chance representations, so the fixed reservoir appears the most channel-robust (raw retention 99%) only because it operates close to chance; an above-chance metric removes most of this advantage. Second, a *zero-imputation* confound: zeroing dropped channels injects out-of-distribution values that penalize non-centered classical features. We prove this metric is not invariant to an information-preserving re-centering, and show empirically—across 10–50% dropout and mean, kNN, and spatial imputation—that any in-distribution fill raises the non-centered encoders by up to 0.11 balanced accuracy and *reverses* the fixed-encoder ranking (ERP-window overtakes the reservoir), while the reservoir's zero-centered embedding is invariant, as predicted. Using the training-free reservoir as an accuracy- and imputation-invariant reference, we show that a trained EEGNet's dropout fragility (0.554 balanced accuracy, 78% retention) is largely a training choice: with channel-dropout augmentation it recovers to 0.674 (96% retention) while remaining the most accurate model (0.70–0.71). We recommend reporting above-chance retention, the channel-imputation rule, and the augmentation state—each a one-line change that materially affects rankings.**

***Index Terms*—EEG, event-related potentials, spiking neural networks, reservoir computing, biomedical signal processing, affective computing, perturbation analysis.**

I. Introduction

Electroencephalographic event-related potentials (ERPs) are time-locked measurements of neural response dynamics that offer millisecond-scale temporal information for biomedical data analysis, yet the observed waveform is a noisy, low-dimensional, and subject-dependent projection of an unobserved neural dynamical system [12], [13]. Inter-subject variability, trial averaging, volume conduction, and measurement noise can cause endpoint classifiers to conflate genuine temporal structure with subject-specific nuisance variation, so a representation that performs well on one clean split may not preserve the same signal properties when amplitude noise, temporal misalignment, or channel loss is introduced.

Reporting only endpoint accuracy does not reveal *which* temporal features a representation preserves or suppresses. This gap is acute for spiking encodings: fixed recurrent reservoirs and spiking networks are expressive nonlinear temporal expansions [1], [4], but in EEG/ERP work they are most often evaluated as end-to-end classifiers [9], [15] rather than characterized as temporal operators whose discrete spike-count observables remain traceable to post-stimulus time windows.

The perturbation behavior of affective-ERP reservoir encodings—their response to amplitude noise, temporal jitter, and channel dropout—is rarely reported, even though preprocessing and acquisition choices materially affect EEG representations [16]–[18].

This paper asks whether reported channel-dropout robustness rankings of affective-ERP representations survive scrutiny of *how* robustness is measured. We audit four representations spanning a fixed-to-trained spectrum—spectral band-power, ERP-window amplitudes, a fixed (untrained) leaky integrate-and-fire (LIF) reservoir summarized by six post-onset spike-count bins (BSC_6) and compressed for a linear readout, and a trained EEGNet (Fig. 1)—and isolate two evaluation confounds that distort the comparison. The fixed, training-free reservoir serves as an accuracy-invariant reference. Graph topology, structure–function coupling, and closed-loop analyses belong to other parts of a broader doctoral framework and are outside this submission. The contributions are:

1) A channel-dropout and amplitude-noise robustness audit of affective-ERP representations spanning fixed and trained encoders, under subject-grouped validation with per-fold confidence intervals and a permutation null (Sec. V-A, Tables II–III).
2) Identification and quantification of two confounds in channel-dropout robustness evaluation: an *accuracy-floor* confound, corrected by an above-chance retention metric, and a *zero-imputation* confound, corrected by train-mean imputation (Sec. III-E, Sec. V-B).
3) A demonstration that the naive robustness ranking *reverses* under the corrected protocol, and that the channel-dropout fragility of a trained EEGNet is largely removed by standard augmentation—separating robustness

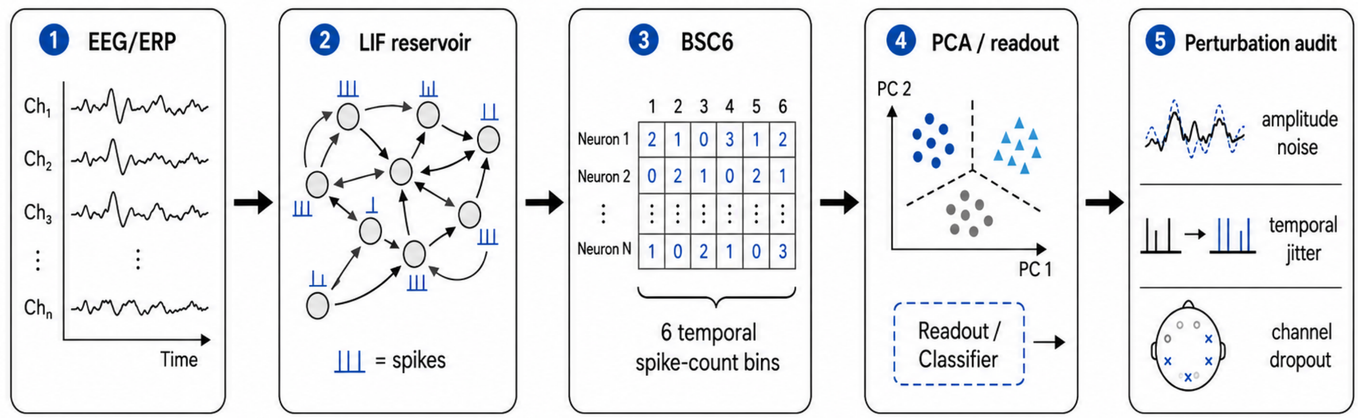

Fig. 1. Reservoir encoding and perturbation-audit pipeline. Trial-averaged affective EEG/ERP observations are mapped through a fixed LIF reservoir, summarized by six temporal spike-count bins (BSC_6), compressed for linear readout, and evaluated under amplitude noise, temporal jitter, and channel-dropout perturbations.

contributed by the representation from robustness contributed by training (Fig. 3, Table III).

4) Use of a fixed, training-free reservoir as an accuracy-invariant reference whose zero-centered embedding is imputation-invariant, which exposes the confound in non-centered classical features; a single-bin ablation isolates the reservoir expansion from temporal bin granularity (Eq. (4)).

Clinical labels and affective categories define the external task structure; the representations are the object of study.

II. Related Work

ERP and EEG analysis have traditionally relied on time-window amplitudes and latencies, time-domain descriptors such as Hjorth parameters [10], spectral and wavelet summaries [11], and linear statistical models [13]. These features are interpretable and correspond to known components such as the P300 and late positive potential, and source-level analyses show that evoked responses carry structured temporal dynamics [12]. Fixed handcrafted windows may, however, underrepresent nonlinear temporal interactions and can be sensitive to subject-specific variability.

Reservoir computing offers an alternative in which a high-dimensional recurrent dynamical system transforms an input sequence into a separable state trajectory [1]–[4]. In liquid state machines, recurrent spiking dynamics induce a nonlinear temporal basis on which a simple readout is trained; related spiking models encode information in spike timing and counts [5], [6]. This separation of fixed dynamics from a trained readout is attractive for biomedical signal analysis because the reservoir can be studied as a measurement operator rather than as a fully optimized black-box classifier.

Spiking and deep networks have been applied to EEG and ERP signals [9], [14], [15], but most studies emphasize endpoint accuracy, while preprocessing and acquisition choices that strongly affect EEG representations are often underexamined [18]. For perturbation-sensitive physiological data, accuracy alone is insufficient: a representation should also be assessed through subject-level generalization, controlled corruption, and representation-geometry analysis [19]. Prior work in the ARSPI-Net line developed neuromorphic reservoir feature extraction for affective EEG [7], [8].

Robustness of EEG models to noise, artifacts, and electrode loss is increasingly evaluated. Dynamic spatial filtering handles missing or corrupted channels with a learned channel-attention front end [23], and channel-dropout and amplitude-noise augmentation are standard ways to harden trained models [24]. A recent benchmark of EEG foundation models reports that no single model is most robust to all perturbations and that much of the apparent channel-dropout fragility disappears when dropped channels are removed rather than zero-padded [22]—showing that perturbation *methodology* shapes the conclusions. We extend this observation in ways the prior work does not: we identify an *accuracy-floor* confound (relative-retention metrics reward low-accuracy representations), quantify a *zero-imputation* confound for fixed-feature representations, and use a fixed, training-free reservoir as an accuracy-invariant reference that separates robustness contributed by the representation from robustness contributed by training. We then show that the naive ranking of channel-dropout robustness reverses once these confounds are corrected.

III. Methods

A. ERP Observation Model

Let $X_s^{(c)} \in \mathbb{R}^{C\times T}$ denote the trial-averaged ERP observation for subject s and affective condition c, with C electrodes (channels) and T temporal samples; observations are stored as $T\times C$ arrays and transposed before encoding. We treat $X_s^{(c)}$ as a noisy observation of an unobserved neural state trajectory:

$$\xi_{t+1} = f_\theta(\xi_t, u_t) + \eta_t, \qquad x_t = H\xi_t + \epsilon_t, \tag{1}$$

where ξ_t is the latent neural state, H is a measurement operator induced by scalp observation, and ϵ_t captures measurement and preprocessing noise. The perturbations below act directly on this observation model (Sec. V-B).

B. Spiking Reservoir Encoding

For each electrode trace, the signal is passed through a fixed LIF reservoir of 256 neurons with recurrent weights W scaled to spectral radius $\rho = 0.9$ (near the edge of stability), input weights W_{in}, leak parameter $\beta = 0.05$, and firing threshold ϑ. A compact discrete-time representation is

$$v_{t+1} = \beta v_t + W z_t + W_{\text{in}} x_t - \vartheta r_t, \tag{2}$$

$$z_{t+1} = \mathbf{1}[v_{t+1} \geq \vartheta], \tag{3}$$

where v_t is the membrane state, z_t is the binary spike vector, and r_t denotes reset after threshold crossing. The reservoir is fixed during feature extraction; only the downstream readout is trained.

Post-onset activity is summarized by six equal temporal bins over the window $t \in [10, 70]$ in sample units, which at 256 Hz spans approximately 39–273 ms post-stimulus. For reservoir unit j and bin k, the binned spike-count code is $b_{j,k} = \sum_{t \in \mathcal{B}_k} z_j(t)$, $k = 1, \ldots, 6$. Per-electrode spike-bin features are compressed to 64 principal components and concatenated across the 34 channels into a trial-level reservoir representation $E_s^{(c)} \in \mathbb{R}^{2176}$. Writing $s_t = z_t \in \{0,1\}^N$ for the reservoir spike state, the encoding is a fixed composition of a nonlinear temporal map, spike-count binning, and the unsupervised projection:

$$E = \phi_{\text{PCA}}\big([b_1, \ldots, b_6]\big) = \Phi(X), \tag{4}$$

so that $E = \Phi(X)$ is a fixed, label-free observable of the input. The principal-component projection is estimated once, unsupervised and without affective labels, over the spike-bin vectors pooled across all observations and electrodes; because the basis is fitted on every subject—including those later held out—it is transductive with respect to subject inclusion, and we report it as a fixed representation audit rather than leakage-free preprocessing (bounded by the fold-local control in Sec. V-A). As a temporal-structure control, we also form a single-bin code (BSC_1, the total post-onset spike count per unit) processed through the identical PCA pipeline. This window is an early, compact interval that excludes the later P300/LPP ranges available to the ERP-window baseline (Sec. III); the clean-classification comparison is therefore conservative for the reservoir.

C. Baselines and Readout

Two classical baselines are evaluated under the same subject-grouped protocol. The band-power baseline (A0) uses spectral band-power features (5 bands $\times$ 34 channels $=$ 170 dimensions). The ERP-window baseline (A0_e) uses mean amplitude in three post-stimulus windows—early (80–200 ms), P300 (250–450 ms), and late positive potential (450–800 ms)—per channel ($3 \times 34 = 102$ dimensions). To separate representational robustness from raw dimensionality, we add a dimensionality-matched control: A0 randomly projected to 2176 dimensions (Gaussian random projection, averaged over five random seeds). The band-power, ERP-window, and reservoir representations use a balanced L2-regularized logistic readout to test separability without a high-capacity classifier.

As a trained deep baseline we evaluate EEGNet-8,2 [14] on the raw 34×256 ERP signal under the identical subject-grouped folds, trained per fold (Adam, 80 epochs) with train-only per-channel standardization. EEGNet ingests the raw signal and has no intermediate feature vector, so its perturbations are applied at the input—comparable to the reservoir's raw-signal recomputation (Sec. V-B)—whereas the band-power, ERP-window, and reservoir perturbations are feature-level; the cross-family comparison is therefore approximate.

D. Perturbation Diagnostics

Two perturbation regimes are separated. Representation-level perturbations corrupt the extracted feature vector; raw-signal recomputation applies perturbations before feature extraction and recomputes the reservoir representation, propagating corruption through reservoir dynamics. Formally, a perturbation $\mathcal{P}_\eta$ acts either on the representation, $E \mapsto \mathcal{P}_\eta(E)$, or on the observation model of Eq. (1): amplitude noise augments the measurement noise ϵ_t, channel dropout zeroes rows of the measurement operator H, and temporal jitter shifts the sampling index of x_t. We apply additive amplitude noise (signal-to-noise ratios), temporal jitter of the alignment window, and channel dropout at increasing electrode-removal rates. Performance is balanced accuracy (BA) and macro-F1.

E. Robustness Metrics and Two Confounds

We report two robustness summaries and isolate two confounds. *Raw retention* is the ratio of perturbed to clean BA. Because BA is bounded below by chance ($c = 1/3$ for three balanced classes), raw retention rewards representations that operate near chance; we therefore also report *above-chance retention*, $(\text{BA}_{\text{pert}} - c)/(\text{BA}_{\text{clean}} - c)$, which removes this accuracy-floor confound. For channel dropout, a dropped channel's features must be filled. The common convention zeroes them, which after train-only standardization maps a feature to $-\mu/\sigma$—an out-of-distribution value whose magnitude grows with the feature mean. We compare this *zero-imputation* with *train-mean imputation*, which maps dropped features to a neutral, in-distribution value; the gap isolates the imputation confound. We also evaluate two further in-distribution fills—k-nearest-neighbor imputation (train-fitted, k=10) and a within-sample spatial fill (the mean of the retained channels)—to check that the effect is not specific to mean imputation. For the trained EEGNet, whose input is the raw signal, we additionally train a variant with channel-dropout and amplitude-noise augmentation, separating robustness contributed by training from robustness contributed by the representation.

Non-invariance of zero-imputation robustness. The imputation confound is a deterministic property of the metric, not of the data. For a standardized feature with training mean μ and standard deviation σ, a dropped value imputed as v enters the

TABLE I
DATASET AND EVALUATION PROTOCOL SUMMARY.

Item	Value
Subjects (after exclusion)	211
Observations	633 (subject $\times$ condition)
Conditions	3 (neg., neutral, pleasant)
Channels / rate / samples	34 / 256 Hz / 256
Band-power (A0) dim.	170 (5$\times$34)
ERP-window (A0$_e$) dim.	102 (3$\times$34)
Reservoir embedding (E) dim.	2176 (64$\times$34)
Validation	subject-grouped 5-fold CV

readout as $(v-\mu)/\sigma$, so zero- and mean-imputation differ by a fixed shift,

$$\frac{0-\mu}{\sigma} - \frac{\mu-\mu}{\sigma} = -\frac{\mu}{\sigma}. \quad (5)$$

Train-only standardization makes the readout invariant to a constant feature offset, so re-centering a representation ($x \mapsto x - b$) leaves its clean accuracy unchanged yet shifts its zero-imputation dropout response by b/σ. Zero-imputation robustness is thus *not invariant* to an information-preserving re-centering: a representation can be made to look arbitrarily more or less channel-robust without changing what it encodes. The confound vanishes exactly when $\mu = 0$, which is why a mean-centered embedding (e.g., PCA features) is a fixed point of the imputation choice and serves as an invariant reference.

IV. Experimental Protocol

A. Dataset and Task

The experiments use a trial-averaged affective ERP dataset with 211 subjects after exclusion of one anomalous subject record. Each subject contributes three affective conditions (negative, neutral, pleasant), yielding 633 subject-condition observations. The ERP traces contain 34 scalp channels down-sampled to 256 Hz, with 256 post-stimulus samples per epoch, baseline-corrected relative to onset. Subject-level data are maintained in a restricted research environment; only aggregate, de-identified results are reported. Table I summarizes the protocol.

B. Validation

All results are computed under a single protocol: StratifiedGroupKFold (5 folds, seeds 42–46), train-only standardization, and a balanced L2 logistic readout; observations from the same subject never span train and test. We report means with 95% confidence intervals (CIs) over the 25 folds, and macro one-vs-rest AUC from pooled out-of-fold probabilities. Chance is calibrated by a label-permutation null (100 permutations).

V. Results

A. Clean Classification and Controls

Table II reports clean three-class performance, and Fig. 2 shows the full accuracy landscape under clean, amplitude-noise, and (naive, zero-imputed) channel-dropout conditions. The trained EEGNet is the most accurate representation (BA 0.711); among the fixed encoders, ERP-window features lead (A0$_e$, 0.616), above band-power (A0, 0.485) and the reservoir embedding (A1, 0.463). A label-permutation null gives BA 0.338 ± 0.022, so every configuration encodes genuine signal (all exceed the null by more than five standard deviations) while the absolute clean separability of the fixed encoders remains modest.

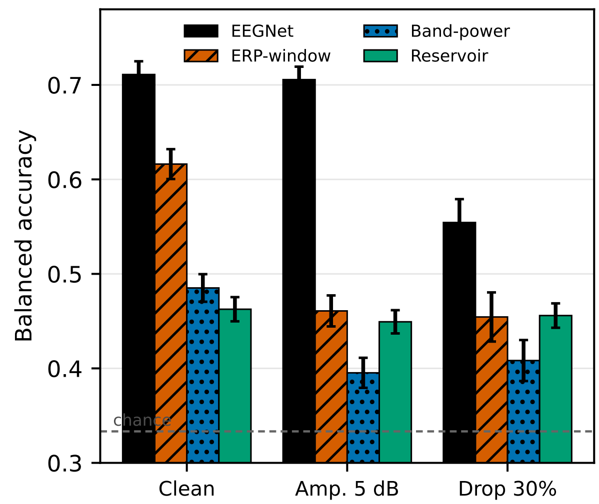

Fig. 2. Accuracy landscape under clean, 5 dB amplitude-noise, and 30% channel-dropout conditions (mean, 95% CI over 25 folds; dropout uses the naive zero-imputation). EEGNet is most accurate throughout; among the fixed encoders the reservoir is least accurate but least degraded, ERP-window strongest on clean data. Sec. V-B shows that the dropout ordering here is an evaluation artifact.

TABLE II
CLEAN SUBJECT-LEVEL PERFORMANCE (MEAN $\pm$ 95% CI OVER 25 FOLDS). PERMUTATION-NULL BA $= 0.338 \pm 0.022$.

Cfg	Features	BA	F1	AUC
D0	EEGNet (raw, trained)	0.711 ± 0.014	0.710	0.868
A0$_e$	ERP-window (102)	0.616 ± 0.016	0.615	0.790
A0	Band-power (170)	0.485 ± 0.015	0.483	0.665
A1	Reservoir BSC$_6$ (2176)	0.463 ± 0.013	0.460	0.644
–	Reservoir BSC$_1$ (2176)	0.467 ± 0.011	0.465	0.646

Temporal-binning ablation. The single-bin total-count code (BSC$_1$) matches BSC$_6$ on clean classification (0.467 vs. 0.463; Table II) and under perturbation, so the reservoir's behavior is driven by the fixed state-space expansion, not by temporal bin granularity.

Transduction and component controls. A fold-local PCA refit (per-fold training subjects) leaves reservoir performance unchanged (BA $0.463 \to 0.465$, AUC $0.644 \to 0.649$), and varying the component count over $\{16, 32, 64, 128\}$ keeps BA within 0.461–0.491, so neither the transductive basis nor the 64-component choice inflates results.

B. Two Confounds in Channel-Dropout Robustness

Under *amplitude* noise the accuracy ranking is preserved and degradation is mild for the high-capacity and reservoir representations (EEGNet $0.711 \to 0.706$, 99% retention; reservoir $0.463 \to 0.449$, 97%), while the classical features fall more (ERP-window $0.616 \to 0.461$; band-power $0.485 \to 0.395$). *Channel dropout* is where evaluation choices dominate the conclusion (Table III, Fig. 3).

Accuracy-floor confound. Under the standard zero-imputation, raw retention ranks the reservoir first by a wide margin (99% at 30% dropout)—but only because it operates near chance, leaving little distance from its clean BA (0.463) to the $1/3$ floor. Above-chance retention removes this artifact and compresses the gap (reservoir 95%, band-power 64%, EEGNet 59%, ERP-window 55%), so a "most robust" verdict from raw retention is largely an accuracy-floor effect.

Zero-imputation confound. Zeroing a dropped channel's features and then standardizing maps them to $-\mu/\sigma$, an out-of-distribution value for features with non-zero mean. Train-mean imputation instead maps them to a neutral value and raises the dropout BA of the classical encoders (band-power $0.408 \to 0.431$, ERP-window $0.454 \to 0.489$ at 30%; up to $+0.043$ at 50%), while the reservoir—whose PCA embedding is zero-centered—is essentially unchanged ($0.456 \to 0.461$). The fixed-encoder dropout ranking therefore *reverses*: zero-imputation ranks the reservoir first ($0.456 > 0.454 > 0.408$), but the corrected imputation ranks ERP-window first ($0.489 > 0.461 > 0.431$). The reservoir's imputation-invariance is exactly what makes it a useful accuracy-invariant reference, isolating the artifact that affects the other encoders.

Generality. The mean-minus-zero gap predicted by Eq. (5) is $+0.034$ (ERP-window), $+0.023$ (band-power), and only $+0.005$ (reservoir) at 30% dropout: it tracks feature non-centeredness and vanishes for the PCA-centered embedding, as predicted. The effect is not specific to mean imputation or to one dropout rate (Fig. 3). Across 10–50% dropout every in-distribution fill (mean, kNN, spatial) raises the non-centered encoders—by up to 0.11 BA for ERP-window under kNN—while leaving the reservoir within 0.04 BA of its zero-imputation value. Consequently the reservoir's naive lead does not survive: at 30% dropout ERP-window significantly exceeds the reservoir under kNN (0.562 vs. 0.491, $p{=}2 \times 10^{-5}$) and spatial fills (0.516 vs. 0.462, $p{=}10^{-3}$) and matches it under mean imputation ($p{=}0.06$), whereas under zero-imputation the two are statistically tied ($p{=}0.8$).

Training confound. The trained EEGNet's apparent dropout fragility (0.554, 78% raw retention) is largely a training choice. Re-training with channel-dropout and amplitude-noise augmentation recovers dropout BA to 0.674 (from 0.554; 96% retention) while preserving clean accuracy (0.703), so for a trained model robustness is contributed mainly by training rather than by the representation (Table III, +aug. row).

Other controls. A dimensionality-matched random projection of band-power (2176-D, five seeds) gives clean BA 0.501 ± 0.002 but collapses to 0.399 ± 0.008 at 5 dB, tracking band-power rather than the reservoir, so the reservoir's robustness is not a dimensionality effect.

Input-level check (reservoir). Recomputing the embedding from corrupted *signals* (perturbation before extraction, as for

TABLE III
CHANNEL DROPOUT AT 30%: NAIVE VS. CORRECTED EVALUATION. "ZERO"/"MEAN" ARE ZERO- AND TRAIN-MEAN IMPUTATION OF DROPPED CHANNELS; RAW-RET. $= \mathrm{BA}_{\mathrm{zero}}/\mathrm{BA}_{\mathrm{clean}}$ (NAIVE); AC-RET. = ABOVE-CHANCE RETENTION UNDER MEAN-IMPUTATION (CORRECTED). EEGNET IS PERTURBED AT ITS RAW (NEAR-ZERO-MEAN) INPUT, SO IMPUTATION IS IMMATERIAL; "+AUG." ADDS TRAIN-TIME AUGMENTATION.

Representation	Clean	Zero	Mean	raw-ret.	AC-ret.
EEGNet	0.711	0.554	–	78%	59%
+aug.	0.703	0.674	–	96%	92%
ERP-window	0.616	0.454	0.489	74%	55%
Band-power	0.485	0.408	0.431	84%	64%
Reservoir	0.463	0.456	0.461	99%	98%

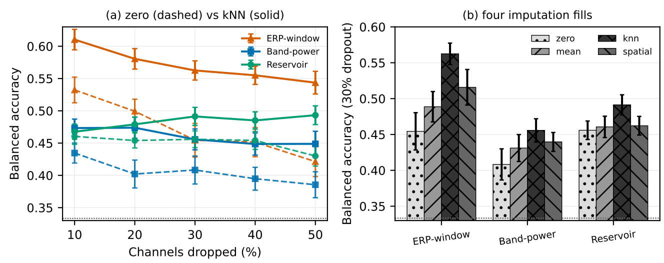

Fig. 3. The imputation confound generalizes across dropout rates and fills. (a) Dropout-rate curves under zero (dashed) and kNN (solid) imputation: the reservoir's zero-centered embedding is imputation-invariant (its curves overlap), whereas the non-centered ERP-window and band-power encoders gain up to 0.11 BA from an in-distribution fill, and kNN-imputed ERP-window overtakes the reservoir at every rate. (b) At 30% dropout under four fills (zero, mean, kNN, spatial), the measured robustness of the non-centered encoders swings by 0.05–0.11 BA with the imputation choice while the reservoir barely moves. Dashed line = chance.

EEGNet) leaves it stable under amplitude noise and dropout (BA 0.449, 0.431 vs. 0.465 clean) but sensitive to temporal jitter (0.422 at ± 50 ms), consistent with its dependence on input timing.

VI. DISCUSSION

Channel-dropout robustness comparisons of EEG representations are dominated by two evaluation choices and, for trained models, by augmentation. Raw retention rewards low-accuracy representations through an accuracy-floor effect; zero-imputation of dropped channels injects out-of-distribution values that penalize non-centered features; and augmentation, not the representation, supplies most of a trained model's dropout robustness. Correcting these factors changes the conclusion: the naive protocol declares the near-chance fixed reservoir the most channel-robust representation, whereas the corrected protocol ranks ERP-window features highest among the fixed encoders and yields an augmented EEGNet that is simultaneously the most accurate and the most channel-robust. The fixed, training-free reservoir is valuable here precisely because it is accuracy- and imputation-invariant: it is a reference that exposes the artifacts affecting the other encoders, not a competitive classifier. A single-bin ablation (BSC_1 matches BSC_6) and a dimensionality-matched random projection further show the reservoir's behavior is driven by the fixed state-space expansion, not by bin granularity or dimensionality.

These findings motivate a minimal reporting protocol for EEG robustness studies: report above-chance retention alongside raw retention; state and vary the channel-imputation rule (zero vs. mean or removal); and, for trained models, report robustness with and without augmentation. Each is a one-line change, yet each materially affects the ranking.

Several boundaries remain. Results are from a single affective-ERP cohort; however, the imputation confound follows from a deterministic property of standardized readouts (Eq. (5)) rather than from the data, and it held across three in-distribution imputation rules and five dropout rates, so we expect it to transfer—external multi-cohort validation is nonetheless the key next step. EEGNet is perturbed at its raw input while the fixed-feature comparisons are feature-level, so the cross-family comparison is approximate, though the input-level reservoir check (Sec. V-B) and the augmentation analysis bracket this gap. We study amplitude noise and channel dropout; adversarial attacks and montage shifts are future work; graph topology and structure–function coupling are out of scope.

VII. Conclusion

Channel-dropout robustness rankings of affective-ERP representations are confounded by the accuracy floor, by zero-imputation of dropped channels, and—for trained models—by the absence of augmentation. Using a fixed, training-free spiking reservoir as an accuracy- and imputation-invariant reference, we showed that correcting these confounds reverses the naive ranking: the reservoir's apparent channel-robustness advantage is largely an evaluation artifact, ERP-window features become the most robust fixed encoder under corrected imputation, and a trained EEGNet with standard augmentation is both the most accurate and the most channel-robust. We recommend reporting above-chance retention, the channel-imputation rule, and the augmentation state in EEG robustness studies. Future work extends the audit to multi-cohort data and additional perturbation families.

Data Availability and Scope. The EEG/ERP data used in this study are access-controlled research data and are not redistributed with this submission. Analyses are reported only as aggregate, de-identified results. Clinical labels, where present in the source cohort, are used only to define biomedical signal-analysis context and are not interpreted as diagnostic decisions.

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